MILITARY SPECIFICATION

CYLINDERS, SAMPLING

1. SCOPE

1.1 Scope. This specification covers a variety of stainless steel cylinders intended primarily for sampling, transportation, and analysis of rocket propellants as compressed gases, liquefied compressed gases, or liquids.

1.2 Classification. Cylinders covered by this specification shall be of the following types, styles, ratings, and sizes (6.2).

Type I -- cylinders with valves and carrying case.
Type II - cylinders only (without valves or carrying case).

Style 1 - double valve connection without safety relief device.
Style 2 - double valve connection with safety relief device.
Style 3 - single valve connection without safety relief device.
Style 4 - single valve connection with safety relief device.

Rating -- high pressure (1800 psi max. service pressure).
Rating -- low pressure (400 or 450 psi max. service pressure).

Size - 10 ml.
Size - 30 ml.
Size - 75 ml.
Size - 150 ml.
Size - 300 ml.
Size - 500 ml.
Size - 1000 ml.
Size - 2250 ml.
Size - 3000 ml.
Size - 1 gal.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601 Boxes, Wood, Cleated-Plywood

FSC 8120

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PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard

Military

MIL-L-25567	Leak Detection Compound, Oxygen Systems
MIL-T-27602	Trichloroethylene, Oxygen Propellant Compatible
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, with Dispenser

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-731	Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Transportation (DOT)

49 CFR 171-190	Hazardous Materials Regulations of the Department of Transportation

National Bureau of Standards (NBS)

Handbook H-28	Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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American Trucking Associations (ATA)

National Motor Freight Classification Rules

(Applications for copies should be addressed to the American Trucking Associations, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Description. Type I cylinders shall consist of an approved Department of Transportation (DOT) specification cylinder with valves and carrying case as specified herein. Type II cylinders shall consist of an approved DOT specification cylinder only as specified herein.

3.2 Cylinders. Cylinders shall conform to DOT specifications 3A, 3B, or 3E and as specified herein. The cylinders shall be true of form: that is, cylindrical and straight with reference to the longitudinal axis within 0.10 inch tolerance, and shall be free of leaks.

3.2.1 Material. The material shall be equivalent in chemical composition to austenitic stainless steel, Type 304 with a molybdenum content not in excess of 0.30 percent.

3.2.2 Pressure rating. The high pressure rating shall be 1800 psi (pounds per square inch) maximum service pressure. The low pressure rating shall be 400 psi or 450 psi maximum service pressure.

3.2.3 Capacity. The volumetric capacity shall be not less than the specified size expressed in milliliters or gallons.

3.2.4 Dimensions. The dimensions for the cylinders including necks and diameters shall be not over those shown in Table I.

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Table I. Cylinder Dimensions

Size	Length (inches max.)	Outside Diameter (inches max.)	Opening Size (NGT) or (NPT)
10 ml	4 1/4	3/4	1/8
30 ml	5.0	1.0	1/8
75 ml	5 3/4	1 1/2	1/4
150 ml	9.0	1 1/2	1/4
300 ml	9 3/4	2.0	3/8
500 ml	14 3/4	2.0	3/8
1000 ml	10 1/4	4.0	1/2
2250 ml	16 3/4	4.0	1/2
3000 ml	21.0	4.0	1/2
1 gal	26.0	4.0	1/2

3.2.5 Openings. The openings (ports) in the cylinder shall be the sizes shown in Table I, threaded with an internal National Gas Taper Thread (NGT) or National Pipe Thread (NPT) conforming to Handbook H-28.

3.2.6 Cleaning. The cylinder shall be cleaned, pickled and passivated internally and externally, and shall be free of any pitting, scale, deposits or oxidation coatings exclusive of the passivation. The internal surfaces shall be free of any oil film.

3.2.7 Delivery date. Cylinders shall be delivered within one year of the test date on the cylinders.

3.2.8 Sealing (Type II only). Ports in the cylinders shall be sealed with shipping plugs.

3.3 Valves (Type I only). Unless otherwise specified, the valves shall be globe or right angle pattern with blunt nosed needle type stems (6.2). The valves shall be capable of sealing to pressure and vacuum and of repeated disassembly for cleaning without special tools or damage to parts.

3.3.1 Material. All metal parts except for the handle and its retaining nut shall be equivalent to austenitic stainless steel Types 304, 304L, or 347.

3.3.2 Packing. Unless otherwise specified, the packing shall be exclusively a polytetrafluoroethylene material (6.2).

3.3.3 Seating surfaces. Unless otherwise specified, the stem and seat seating surfaces shall be stainless steel as specified in 3.3.1 (6.2).

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3.3.4 Safety relief device. When required, the safety relief device shall conform to DQT requirements and shall be an integral part of the valve.

3.3.5 End fittings. The valve end fittings shall be integral to the valve. The end which mates to the cylinder shall be male configuration. The threads shall be National Gas Taper Thread (NGT) or National Pipe Threads (NPT) conforming to Handbook H-28 and the sizes shown in Table I.

3.3.6 Handle. The handle shall be not less than 2 inches in diameter or across the legs and shall permit a firm grip by an average sized hand.

3.3.7 Lubrication. Unless otherwise specified, the valve shall be assembled with no lubrication (dry) (6.2).

3.4 Cylinder assembly (Type I only). When assembling the valve to the cylinder, the valve port which sees the bottom of the stem shall be mated to the cylinder. The valve port which sees the upper stem and packing area shall be sealed in accordance with 3.4.5.

3.4.1 Orientation. When two valves are specified, the valve handles or the exit ports, whichever is applicable, shall be oriented in a like direction with respect to the cylinder.

3.4.2 Thread sealant. Unless otherwise specified, the thread sealant shall conform to MIL-T-27730 (6.2). The sealant tape shall be applied in such a manner that the first complete or lead thread remains dry.

3.4.3 Leak test. The assembled cylinder shall be leak tested in accordance with paragraph 4.5.

3.4.4 Cleaning and drying. The assembled and tested cylinder shall be cleaned by thoroughly flushing with trichloroethylene conforming to MIL-T-27602 or equivalent, and dried with oil-free compressed air or nitrogen, which is filtered through a 40 micron absolute filter and has a dew point of -65°F or less.

3.4.5 Seals. Stainless steel pressure caps or plugs conforming to the materials in 3.3.1 shall be installed on the open ends of the valves. The caps or plugs shall be capable of safely retaining the service pressure without failure or leakage.

3.5 Carrying case (Type I only). The carrying case shall conform to PPP-B-621, Style 6 (lock corner), and to the additional requirements specified herein.

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3.5.1 Material. All wood (hardwood) shall conform to MIL-STD-731, Group IV.

3.5.2 Dimensions. The height, width, and length of the carrying case shall not exceed those specified in Table II. The thickness of the wood members shall be not less than those specified in Table II. The cover, sides, and bottom shall be of the same thickness. The ends may be thicker.

Table II. Carrying Case Dimensions

Size	Height & Width maximum (inches)	Length maximum (inches)	Thickness minimum (inches)
10 ml	5	10	1/4
30 ml	5	10	1/4
75 ml	5	15	3/8
150 ml	5	15	3/8
300 ml	6	24	3/8
500 ml	6	24	3/8
1000 ml	8	30	1/2
2250 ml	8	30	1/2
3000 ml	8	40	1/2
1 gal	8	40	1/2

3.5.3 Reinforcements. Steel corner and edge reinforcements located internally or externally or both may be used as deemed necessary.

3.5.4 Cover. The cover shall be hinged and latched.

3.5.4.1 Hinges. The hinges shall be steel butt, offset butt, or piano type. They shall be securely attached to the cover and body in such numbers and manner as to withstand repeated shipment and long term use without loosening.

3.5.4.2 Latches. The latches shall be steel and of a draw-down, positive locking type with provisions for being wired and sealed in the closed position. The positioning and number of latches shall be sufficient to hold the cover firmly closed against the body even during rough handling.

3.5.5 Handles. The handles shall be a heavy duty steel collapsible type. One gravity centered handle on either the cover or one

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side shall be installed on carrying cases with a cylinder of less than 1000 ml capacity. Two handles, one on each end, shall be installed on carrying cases with a cylinder of 1000 ml capacity and above.

3.5.6 Supports. The carrying case shall be provided with wood or metal or wood and metal cylinder assembly supports. They shall be installed as a fixed and integral part of the carrying case. They shall lock the cylinder assembly in a fixed position within the case and prevent the cylinder from falling out when the cover is opened. Cellular cushioning material shall not be used for cylinder assembly support.

3.5.7 Finish. The outside of the carrying case shall be protected with at least one coat of clear varnish or lacquer. The inside may remain unprotected.

3.5.8 Markings. A smooth surfaced white plastic shipping plate with permanent black lettering conforming to figure 1 shall be permanently attached to the cover of the carrying case. The lettering shall be not less than one quarter inch high. Markings specified in PPP-B-621 may be omitted.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection is deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (4.3).
(b) Inspection of preparation for delivery (4.4).
(c) Leak test (4.5).

4.3 Quality conformance inspection.

4.3.1 Lot. A lot shall consist of not more than 200 cylinders offered for delivery at the same time.

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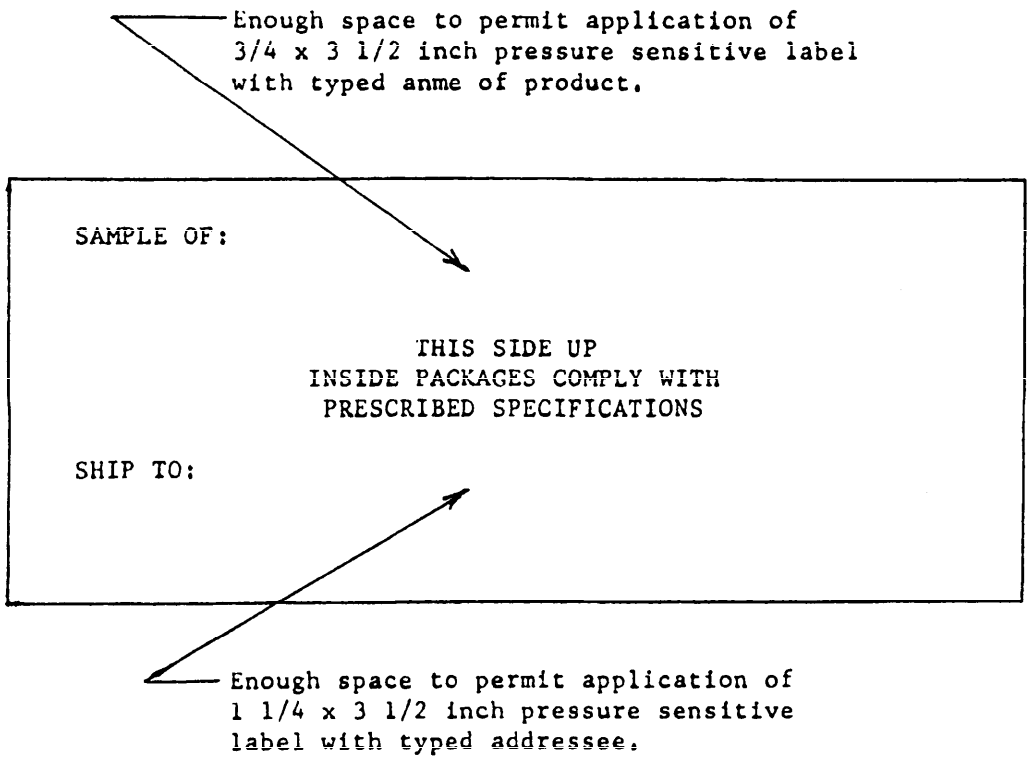


Figure 1. Shipping Plate

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4.3.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined for the following defects using an inspection level of S-2 and an AQL of 1.0 percent defective.

Cylinders (Type I & II).

1.	DoT specification requirements not as specified.	3.2
2.	Material not as specified.	3.2.1
3.	Pressure rating less than specified.	3.2.2
101.	Capacity less than specified.	3.2.3
102.	Dimensions over maximum specified.	3.2.4
103.	Openings wrong size.	3.2.5
104.	Threads incomplete, rough, or torn.	3.2.5
105.	Cylinder not clean as specified.	3.2.6
106.	Test date more than one year prior to delivery date.	3.2.7
107.	Cylinder not plugged (Type II only).	3.2.8

Valves (Type I only).

4.	Valve packing material not as specified.	3.3.2
5.	Valve lubrication not as specified.	3.3.7
6.	Safety relief device not as specified.	3.3.4
108.	Configuration not as specified.	3.3
109.	Material not as specified.	3.3.1
110.	Sealing surfaces not as specified.	3.3.3
111.	Valve ends not integral or size specified.	3.3.5
112.	Threads incomplete, rough, or torn.	3.3.5
113.	Handle not as specified.	3.3.6

Cylinder assembly (Type I only).

7.	Valve mated to cylinder not as specified.	3.4
8.	Thread sealant not as specified.	3.4.2
114.	Valves not oriented as specified.	3.4.1
115.	Cylinder assembly leaks.	3.4.3
116.	Cylinder assembly not clean and dry as specified.	3.4.4
117.	Seals not as specified.	3.4.5

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Carrying case (Type I only).

118.	Carrying case does not meet specification requirements.	3.5
119.	Material not as specified.	3.5.1
120.	Dimensions not within limits.	3.5.2
121.	Reinforcements not used as required.	3.5.3
122.	Cover, hinges, and latches not as specified.	3.5.4
123.	Handle not as specified.	3.5.5
124.	Supports not as specified.	3.5.6
126.	Markings not as specified.	3.5.8
201.	Finish not as specified.	3.5.7

4.4 Inspection of preparation for delivery.

4.4.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.4.2 Sampling. Sampling shall be in accordance with MIL-STD-105.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for the following defects using an inspection level of S-2 and an AQL of 4.0 percent defective.

Delivery pack.

127.	Petroleum-base solvents, preservatives, or lubricants applied to cylinders.	5.1
128.	Cylinders not packed as specified.	5.2
129.	Markings not as specified.	5.3

4.5 Leak test (Type I only). Cylinders shall be tested for leakage. Assembled cylinders shall be charged to the indicated service pressure with oil-free compressed air or nitrogen which is filtered through a 40 micron absolute filter and has a dew point of -65°F or less. The cylinder assembly shall be completely immersed in water, or all joints and packing shall be coated with leak test solution conforming to MIL-L-25567 and observed for bubbles for a minimum of 5 minutes. Any bubbles shall constitute failure of the test. Leakage may be corrected and the cylinder assembly retested. All residual test solution shall be thoroughly washed off.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Petroleum-base solvents, preservatives, and lubricants shall not be applied to any part of the cylinders.

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5.2 Packing. Packing shall be level A, B, or C as specified (6.2).

5.2.1 Level A. All Type I cylinders and Type II cylinders 30 pounds or less gross weight shall be packaged in close-fitting fiberboard boxes conforming to PPP-B-636, overseas type. Type II cylinders over 30 pounds gross weight shall be packaged in close-fitting wooden boxes conforming to PPP-B-601, overseas type, style A or B.

5.2.2 Level B. The cylinders shall be boxed as specified for level A except boxes shall be domestic type.

5.2.3 Level C. The cylinders shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules or other carrier rules applicable to the mode of transportation.

5.3 Marking. Packages shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Cylinders covered by this specification are intended for but not limited to the sampling, storage, and transportation of hazardous rocket propellants as compressed gases, liquefied compressed gases, or liquids. They are not intended for sampling cryogenic compressed gases. Type I cylinders with valves and carrying cases are for sampling and shipment to other points for analysis. Type II cylinders are for analysis and general purpose usage within local areas. The cylinders are frequently disassembled and cleaned before being used for sampling.

6.1.1 When required by other authority, the user is responsible for marking the cylinder with the name of the specific commodity contained with the cylinder and any color coding of the cylinder.

6.1.2 When required by other authority, the user is responsible for attaching to the cylinder an appropriate sample identification tag.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents.

(a) Title, number, and date of this specification.
(b) Type, Style, Rating, and Size (Type I) required (1.2).
(c) When a different valve is required (3.3).
(d) When a different packing material or configuration is required (3.3.2).

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(e) When a different seat configuration or material is required (3.3.3).
(f) When lubrication on valve parts is required (3.3.7).
(g) When a different or no thread lubricant is required (3.4.2).
(h) When a different or no color marking is required (3.4.6).
(i) Level of packing required (5.2).

Custodians:

Air Force - 12

Review Activities:

Air Force - 19, 68, 70

Civilian Agency Interest:

NAS

Preparing Activity:

Air Force - 12

Project No. 8120-F289

Reviewer/user information is current as of the date of this document. For further coordination of changes to this document, draft circulation should be based on the information in the current DoD Index of Specifications and Standards.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-C-83690(USAF) Cylinders, Compressed Gas, Sampling.

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD, FORM 1426 APR 63 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Edwards, California 93523

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